

Heat Watch App Documentation

Overview

Heat Watch is a smart urban planning and climate resilience application designed to visualize heat dynamics, simulate environmental interventions, and promote heat equity within city wards. By integrating satellite data (Landsat API) with real-time analytics, Sunshine empowers users to understand, mitigate, and navigate urban heat islands.

1. Interactive Heat Mapping

The core visualization engine provides real-time thermal analysis overlaid on Google Maps.

- * **Data Source:** Powered by the Landsat API to display accurate temperature gradients.
- * **View Modes:**
 - **Day/Night Toggle:** Switch between daytime peak heat and nighttime retention views.
 - **Zonal Division:** The map is automatically divided based on distinct temperature zones.
- * **User Interface:**
 - **Dynamic Legend:** Color-coded scale ranging from Cool (<35°C) to Extreme (>43°C).
 - **Smart Markers:** Tap any specific zone to reveal detailed temperature metrics and coordinates.

2. Urban Intervention Simulator

A predictive modeling tool that allows users to visualize the impact of green infrastructure before implementation.

- * **Actions:** Users can simulate the following interventions via a drag-and-drop interface:
 - Planting Trees
 - Applying Cool Roofs
 - Adding Parks
- * **Impact Analysis:**
 - **Temperature Reduction:** Visualizes how the local temperature drops in response to interventions.
 - **Energy Savings:** Calculates projected energy savings resulting from reduced cooling loads.
 - **Before/After Comparisons:** Instant side-by-side visual feedback of the landscape changes.
- * **Planning Tools:** Includes a Cost Estimation feature that provides projected budget requirements for the simulated scenarios.

3. Data Analytics & Heat Inequality Index (HII)

A comprehensive dashboard designed to analyze socio-economic factors driving heat vulnerability.

- * Multi-Layer Data: Toggle data overlays to view:
 - Population Density
 - Average Income
 - Vegetation Index (NDVI)
 - Surface Albedo (reflectivity)
- * Heat Inequality Index (HII):
 - Ranks city wards based on a composite Risk Score.
 - Vulnerability Factors: Analyzes Population Density, Green Cover %, Vulnerable Age Groups, and Hospital Access.

4. AI Recommendations

An intelligent advisory engine that suggests specific actions based on the current environmental state of a selected ward.

- * Context Aware: Analyzes real-time data (e.g., low vegetation + high albedo) to propose tailored interventions.
- * Actionable Advice: Provides specific recommendations such as "Increase tree cover by 15% in Ward 4" or "Prioritize cool roofs in industrial zones."

5. Navigation & Historical Analysis

Tools for long-term tracking and immediate citizen utility.

- * Shadiest Path Finder:
 - Input points A and B to calculate the optimal route.
 - Prioritizes paths with maximum shade and tree cover to minimize heat exposure during transit.
- * Historical Heat Maps:
 - Access an archive of past heat maps to track warming trends over months or years.

(Screenshots of the app is included in the next page)









